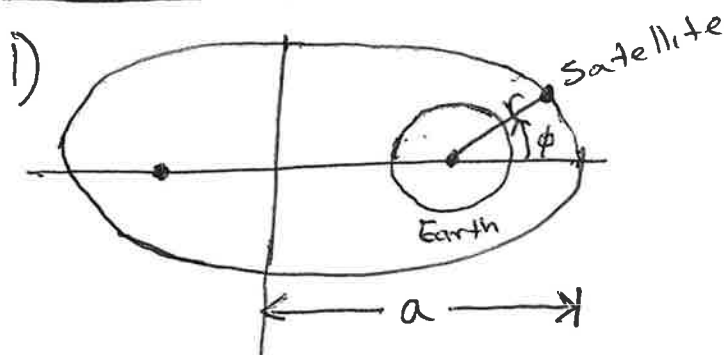


# PHYSICS PROBLEM SET 2 (SOLUTIONS)



Given:  $r_E = 6,371 \text{ km}$

Perigee:  $r_{\min} = r_E + 300 \text{ km}$

Apogee:  $r_{\max} = r_E + 3,500 \text{ km}$

$\Rightarrow r_{\min} = 6,671 \text{ km}$

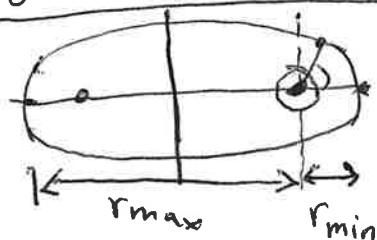
$r_{\max} = 9,871 \text{ km}$

Each part of the problem requires the use of:

$$r(\phi) = \frac{a(1 - \epsilon^2)}{1 + \epsilon \cos \phi} \quad (\text{ellipse w/ origin @ right focus point})$$

but "a" and "ε" must first be determined.

determine semi-major axis (a)



$\Rightarrow 2a = r_{\min} + r_{\max}$

$a = \frac{r_{\min} + r_{\max}}{2} = \underline{\underline{8271 \text{ km}}}$

determine eccentricity ε

Recall:  $\epsilon = \frac{f}{a}$ ,  $f = a - r_{\min} \Rightarrow \epsilon = \frac{a - r_{\min}}{a} = 1 - \frac{r_{\min}}{a}$

$\Rightarrow \underline{\underline{\epsilon = 0.193}}$

Now that we know a & ε, we can use  $r(\phi)$  to solve parts A & B

PART A

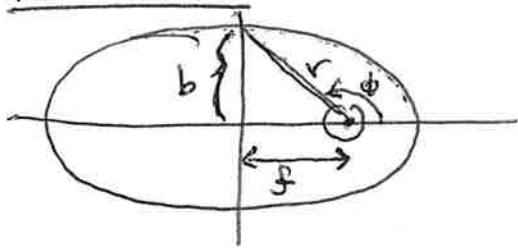
$r(\phi = 90^\circ) = \frac{a(1 - \epsilon^2)}{1 + \epsilon \cos(90^\circ)} = \frac{a(1 - \epsilon^2)}{1} \Rightarrow 7,961 \text{ km}$

$d_{\text{above Earth}}$

$= r(\phi = 90^\circ) - r_E \Rightarrow$

$d_{\text{above Earth}} = \underline{\underline{1590 \text{ km}}}$

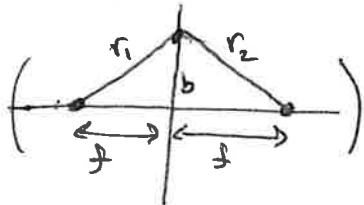
## PART B



$$r^2 = b^2 + f^2$$

$$= b^2 + e^2 a^2$$

To determine  $b$ , recall the definition of ellipse, requires:



$$r_1 + r_2 = 2a$$

or symmetric at  $b$ ,  $r_1 = r_2 = a$

$$\Rightarrow a^2 = b^2 + f^2 = b^2 + e^2 a^2 \Rightarrow b^2 = a^2 - e^2 a^2$$

$$b^2 = a^2(1 - e^2)$$

$\Rightarrow$  By geometry,  $r(\phi) = a$

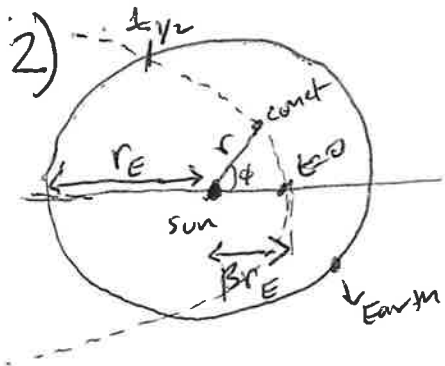
or from the expression  $b^2 = a^2 - e^2 a^2$

$$\text{then } \Rightarrow r^2 = b^2 + e^2 a^2 = a^2 - \cancel{e^2 a^2} + \cancel{e^2 a^2}$$

$$\Rightarrow \boxed{r = a = 8,271 \text{ km}}$$

$$d_{\text{above Earth}} = r - r_E \Rightarrow$$

$$\boxed{d_{\text{above Earth}} = 1900 \text{ km}}$$



$$r(\phi) = \frac{l^2/\mu^2 \bar{\gamma}}{1 + e \cos \phi} \quad \bar{\gamma} = G(m_1 + m_2) \quad e = 1 \text{ (for parabola)}$$

Energy Conservation (for sun + comet)

$$E = \frac{1}{2} \mu \left( \frac{dr}{dt} \right)^2 + V_{\text{eff}}(r), \quad V_{\text{eff}}(r) = \frac{l^2}{2\mu r^2} + V(r)$$

$$\Rightarrow \left( \frac{dr}{dt} \right)^2 = \frac{2}{\mu} [E - V_{\text{eff}}(r)] \Rightarrow \frac{dr}{dt} = \pm \sqrt{\frac{2}{\mu} \left[ E - \frac{l^2}{2\mu r^2} + \frac{\mu}{r} \right]}$$

$$\Rightarrow \frac{dr}{dt} = \pm \sqrt{\frac{2}{\mu} \left[ \frac{\bar{\gamma} \mu}{r} - \frac{l^2}{2\mu r^2} \right]}$$

$$= \pm \sqrt{\frac{2\bar{\gamma}}{r} - \frac{l^2}{\mu^2 r^2}}$$

$$= \pm \sqrt{\frac{2\bar{\gamma} r}{r^2} - \frac{2\beta r_E \bar{\gamma}}{r^2}}$$

$$\Rightarrow \int_0^{t_{1/2}} dt = \int_{r=\beta r_E}^{r_E} \frac{r dr}{\sqrt{2\bar{\gamma} r - 2\beta r_E \bar{\gamma}}} = \frac{1}{\sqrt{2\bar{\gamma}}} \int_{\beta r_E}^{r_E} \frac{r dr}{\sqrt{r - \beta r_E}}$$

$$\int \frac{r dr}{\sqrt{r - \beta r_E}} = \frac{2}{3} \sqrt{r - \beta r_E} (2\beta r_E + r) \Big|_{\beta r_E}^{r_E}$$

$$= \frac{2}{3} \sqrt{r_E - \beta r_E} (2\beta r_E + r_E)$$

$$\Rightarrow 2 \cdot \frac{t_{1/2}}{\sqrt{2\bar{\gamma}}} = \frac{2 \cdot 1}{\sqrt{2\bar{\gamma}}} \cdot \frac{2}{3} \sqrt{r_E - \beta r_E} (2\beta r_E + r_E)$$

time of comet from closest approach to Earth's radius orbit (half-orbit) time

Multiply both sides by  $\sqrt{2\bar{\gamma}}$  to obtain the total time the comet spends in Earth's orbit

$$\frac{2}{\sqrt{2\bar{\gamma}}} \rightarrow \frac{2\sqrt{2}}{\sqrt{2} \cdot \sqrt{2\bar{\gamma}}} = \sqrt{\frac{2}{\bar{\gamma}}}$$

2) continued...

$$t_0 = \sqrt{\frac{2}{\gamma}} \cdot \frac{2}{3} \sqrt{r_E(1-\beta)} (r_E(1+2\beta))$$

$$= \frac{2}{3} \sqrt{\frac{2}{\gamma}} \sqrt{1-\beta} (1+2\beta) \cdot r_E^{3/2} \Rightarrow t_0 \propto r_E^{3/2}$$

Use Kepler's 3rd LAW for Earth's orbit

$$T_E^2 = \frac{4\pi^2 a^3}{\gamma}, \text{ where for Earth's orbit around the sun, } a \rightarrow r_E$$

$$\Rightarrow T_E = \frac{2\pi}{\sqrt{\gamma}} r_E^{3/2} \rightarrow \boxed{r_E^{3/2} = \frac{T_E \cdot \sqrt{\gamma}}{2\pi}}$$

Now substitute  $r_E^{3/2}$  expression in  $t_0$ : where  $T_E = 1 \text{ year}$   
(Earth orbit period)

$$\Rightarrow t_0 = \frac{2}{3} \sqrt{\frac{2}{\gamma}} \sqrt{1-\beta} (1+2\beta) \cdot \frac{\cancel{\sqrt{\gamma}}}{2\pi} \times 1 \text{ year}$$

$$\Rightarrow \boxed{t_0 = \frac{\sqrt{2(1-\beta)} (1+2\beta)}{3\pi} \times T_E}$$

$$3) \quad r(\phi) = k e^{\alpha \phi}, \quad k, \alpha > 0 \text{ (positive const.)}$$

PART A

$$F(r) = -\nabla V(r) = -\frac{dV(r)}{dr} \rightarrow \text{Need to determine } V(r)$$

Use Energy conservation in a central-force field:

$$E = \frac{1}{2} \mu \left( \frac{dr}{dt} \right)^2 + \frac{l^2}{2\mu r^2} + V(r)$$

$$\text{Now, } \frac{dr}{dt} = \frac{dr}{d\phi} \frac{d\phi}{dt}, \text{ Recall: } l = \mu r^2 \frac{d\phi}{dt} \Rightarrow \frac{d\phi}{dt} = \frac{l}{\mu r^2}$$

$$\text{from } r(\phi) = k e^{\alpha \phi} \Rightarrow \frac{dr}{d\phi} = \underline{k e^{\alpha \phi} \cdot \alpha} = r \alpha$$

$$\Rightarrow \frac{dr}{dt} = k \alpha \cdot e^{\alpha \phi} \cdot \frac{l}{\mu r^2} \rightarrow \text{Substitute expression into } E \text{ "energy cons. eq."}$$

$$\Rightarrow V(r) = E - \frac{l^2}{2\mu r^2} - \frac{1}{2} \mu \cdot \left( \underbrace{k^2 \alpha^2 e^{2\alpha \phi}}_{\alpha^2 \cdot \underbrace{k^2 e^{2\alpha \phi}}_{r^2}} \cdot \frac{l^2}{(\mu r^2)^2} \right)$$

$$= E - \frac{l^2}{2\mu r^2} - \frac{1}{2} \frac{\mu \alpha^2 r^2 l^2}{\mu^2 r^4} = E - \frac{l^2}{2\mu r^2} - \frac{\alpha^2 l^2}{2\mu r^2}$$

$$= E - \frac{l^2}{2\mu r^2} - \alpha^2 \frac{l^2}{2\mu r^2} \Rightarrow \boxed{V(r) = E - \frac{l^2}{2\mu r^2} (1 + \alpha^2)}$$

$$\Rightarrow \frac{dV}{dr} = -\frac{l^2}{2\mu} (1 + \alpha^2) \cdot (-2r^{-3}) = \frac{l^2 (1 + \alpha^2)}{\mu r^3}$$

$$\Rightarrow \boxed{F(r) = -\frac{l^2}{\mu r^3} (1 + \alpha^2)}$$

### 3) PART B

determine  $\phi(t)$ : Recall  $\frac{d\phi}{dt} = \frac{l}{\mu r^2} = \frac{l}{\mu k^2 e^{2\alpha\phi}}$

$$\int e^{2\alpha\phi} d\phi = \int \frac{l}{\mu k^2} \cdot dt \rightarrow \frac{e^{2\alpha\phi}}{2\alpha} + C = \frac{lt}{\mu k^2}$$

$$\ln e^{2\alpha\phi} = \ln \left[ \frac{2\alpha lt}{\mu k^2} + C' \right] \Rightarrow 2\alpha\phi = \ln \left[ \frac{2\alpha lt}{\mu k^2} + C' \right]$$

$$\boxed{\phi(t) = \frac{1}{2\alpha} \ln \left( \frac{2\alpha lt}{\mu k^2} + C' \right)}$$

determine  $r(t)$ : Now we can substitute  $\phi(t)$  to  $r(\phi) = k e^{\alpha\phi(t)}$  to obtain  $r(t)$

$$\begin{aligned} \Rightarrow r(t) &= k e^{\alpha \cdot \left( \frac{1}{2\alpha} \ln \left( \frac{2\alpha lt}{\mu k^2} + C' \right) \right)} \\ &= k e^{\ln \left( \frac{2\alpha lt}{\mu k^2} + C' \right)^{1/2}} \end{aligned}$$

$$\Rightarrow r(t) = \sqrt{k^2} \cdot \sqrt{\frac{2\alpha lt}{\mu k^2} + C'}$$

$$\boxed{r(t) = \sqrt{\frac{2\alpha lt}{\mu} + k^2 C'}}$$

### 3) PART C

Recall:  $E = \frac{1}{2} \mu \left( \frac{dr}{dt} \right)^2 + \frac{l^2}{2\mu r^2} + V(r)$

$$\rightarrow \frac{dr}{dt} = \frac{dr}{d\phi} \frac{d\phi}{dt} = r \alpha \cdot \frac{l}{\mu r^2} \rightarrow \left( \frac{dr}{dt} \right)^2 = \frac{\alpha^2 l^2}{\mu^2 r^2}$$

$\rightarrow$  Also recall:  $V(r) = E - \frac{l^2}{2\mu r^2} (1 + \alpha^2)$

$$E = \frac{1}{2} \mu \frac{\alpha^2 l^2}{\mu^2 r^2} + \frac{l^2}{2\mu r^2} + V(r)$$

$$= \frac{\alpha^2 l^2}{2\mu r^2} + \frac{l^2}{2\mu r^2} + V(r) = \frac{l^2}{2\mu r^2} (1 + \alpha^2) + V(r)$$

$$\Rightarrow E = \frac{l^2}{2\mu r^2} (1 + \alpha^2) - \frac{l^2}{2\mu r^2} (1 + \alpha^2) = 0$$

$$4) \quad r(\phi) = k\phi^2, \quad k > 0$$

$$F(r) = -\frac{dV}{dr}, \quad \text{Need to determine } V(r)$$

$$\text{From energy conservation, recall } \Rightarrow E = \frac{1}{2} \mu \left( \frac{dr}{dt} \right)^2 + \frac{l^2}{2\mu r^2} + V(r)$$

$$\Rightarrow V(r) = E - \frac{l^2}{2\mu r^2} - \frac{1}{2} \mu \left( \frac{dr}{dt} \right)^2$$

$$\frac{dr}{dt} = \frac{dr}{d\phi} \frac{d\phi}{dt} \rightarrow \frac{dr}{d\phi} = 2k\phi, \quad \frac{d\phi}{dt} = \frac{l}{\mu r^2} = \frac{l}{\mu k^2 \phi^4}$$

$$\cancel{\mu} (2k\phi) \left( \frac{l}{\mu k^2 \phi^4} \right)^2 = \underbrace{4k^2 \phi^2}_{k(k\phi^2)} \cdot \frac{l^2}{\mu^2 r^4} = 4kr \cdot \frac{l^2}{\mu^2 r^4}$$

$$\Rightarrow \left( \frac{dr}{dt} \right)^2 = \frac{4kl^2}{\mu^2 r^3} \leftarrow \text{substitute into } V(r) \text{ expression}$$

$$V(r) = E - \frac{l^2}{2\mu r^2} - \frac{1}{2} \mu \frac{4kl^2}{\mu^2 r^3}$$

$$V(r) = E - \frac{l^2}{2\mu r^2} - \frac{2kl^2}{\mu r^3}$$

$$F(r) = -\frac{dV}{dr}, \quad \text{Now, } \frac{dV}{dr} = \frac{l^2}{2\mu} (-2r^{-3}) - \frac{2kl^2}{\mu} (-3r^{-4})$$

$$\Rightarrow \frac{dV}{dr} = \left( + \frac{l^2}{\mu r^3} + \frac{6kl^2}{\mu r^4} \right) \quad \cancel{+4}$$

$$\boxed{F(r) = -\frac{l^2}{\mu} \left( \frac{6k}{r^4} + \frac{1}{r^3} \right)}$$

$$5) V(r) = -\frac{k}{r} e^{-r/a}, \quad e^{-r/a} \approx 1 - \frac{r}{a} + \frac{1}{2!} \left(\frac{r}{a}\right)^2 - \dots$$

only keep terms to 1<sup>st</sup> order  $\left(\frac{r}{a}\right)$

$$\Rightarrow V(r) = -\frac{k}{r} \left(1 - \frac{r}{a}\right) = -\frac{k}{r} + \frac{k}{a}$$

To obtain a general  $r(\phi)$ , start with energy conservation:

$$E = \frac{1}{2} \mu \left(\frac{dr}{dt}\right)^2 + \frac{l^2}{2\mu r^2} + V(r), \quad \frac{dr}{dt} = \frac{dr}{d\phi} \underbrace{\frac{d\phi}{dt}}_{\frac{l}{\mu r^2}}$$

$$E = \frac{1}{2} \mu \left(\frac{dr}{d\phi}\right)^2 \frac{l^2}{(\mu r^2)^2} + \frac{l^2}{2\mu r^2} + V(r)$$

$$\Rightarrow \left(\frac{dr}{d\phi}\right)^2 = \sqrt{\frac{2}{\mu} \left[E - \frac{l^2}{2\mu r^2} - V(r)\right]} \cdot \sqrt{\frac{(\mu r^2)^2}{l^2}}$$

$$\Rightarrow d\phi = \frac{l}{\mu r^2} \frac{dr}{\sqrt{\frac{2}{\mu} \left[E - \frac{l^2}{2\mu r^2} - V(r)\right]}}$$

$$\int d\phi = \frac{l}{\sqrt{2\mu}} \int \frac{dr}{r^2 \sqrt{E - \frac{l^2}{2\mu r^2} - V(r)}}$$

$$E - \frac{l^2}{2\mu r^2} + \frac{k}{r} - \frac{k}{a}$$

$$\left(E - \frac{k}{a}\right) + \frac{k}{r} - \frac{l^2}{2\mu r^2}$$

Now let:  $A \equiv -\frac{l^2}{2\mu}$ ,  $B \equiv k$ ,  $C \equiv E - \frac{k}{a}$

and  $U = \frac{1}{r}$   $dU = -\frac{dr}{r^2}$

$$\Rightarrow \phi - \phi_0 = -\frac{l}{\sqrt{2\mu}} \int \frac{dU}{\sqrt{AU^2 + BU + C}}$$

$$= -\frac{l}{\sqrt{2\mu}} \left[ -\frac{1}{\sqrt{-A}} \sin^{-1} \left( \frac{2AU + B}{\sqrt{B^2 - 4AC}} \right) \right]$$

5) continued...

$$\phi - \phi_0 = - \underbrace{\frac{l}{\sqrt{2\mu}} - \frac{\sqrt{2\mu}}{\sqrt{l^2}}}_{\equiv 1} \sin^{-1} \left( \frac{2Au + B}{\sqrt{B^2 - 4AC}} \right)$$

$$\sin(\phi - \phi_0) = \frac{2Au + B}{\sqrt{B^2 - 4AC}}$$

Now solve for  $u = 1/r$ :

$$2Au = \frac{\sqrt{B^2 - 4AC}}{2A} \sin(\phi - \phi_0) - \frac{B}{2A}$$

$$\begin{aligned} \frac{1}{u} &= \frac{2A}{\sqrt{B^2 - 4AC} \sin(\phi - \phi_0) - B} \\ &= \frac{2A}{-B \left( 1 - \frac{\sqrt{B^2 - 4AC}}{B} \sin(\phi - \phi_0) \right)} \end{aligned}$$

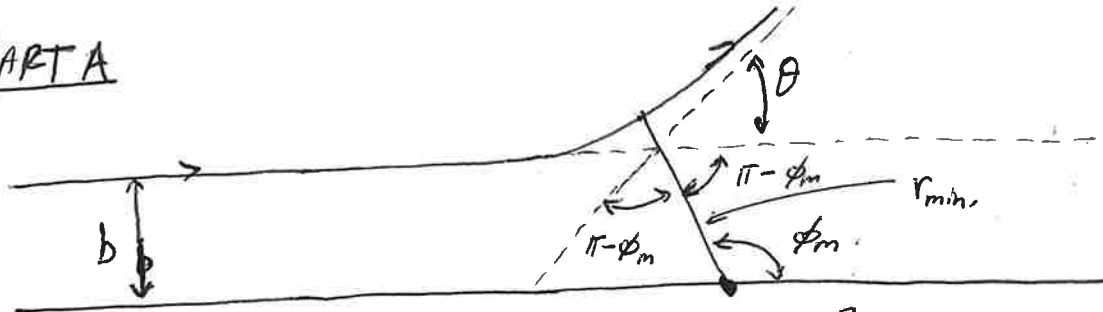
$$\Rightarrow r = \frac{-2A/B}{1 - \frac{\sqrt{B^2 - 4AC}}{B} \sin(\phi - \phi_0)}$$

Now substitute back  $A, B, C$ :

$$\frac{-2A}{B} = -2 \left( -\frac{l^2}{2\mu k} \right) = l^2 / \mu k \equiv \alpha$$

$$\begin{aligned} \frac{\sqrt{B^2 - 4AC}}{B} &= \frac{\sqrt{k^2 + 4 \left( \frac{l^2}{2\mu} \right) \left( E - \frac{k}{a} \right)}}{\sqrt{k^2}} = \sqrt{1 + \frac{2 l^2}{\mu k^2} \left( E - \frac{k}{a} \right)} \equiv \epsilon \\ \Rightarrow r(\phi) &= \frac{l^2 / \mu k}{1 - \epsilon \sin(\phi - \phi_0)} \end{aligned}$$

# 6) PART A



Starting from total energy:  $E = \frac{1}{2} M \dot{r}^2 + \frac{l^2}{2Mr^2} + V(r)$

Use relation:  $\frac{dr}{dt} = \frac{dr}{d\phi} \dot{\phi}$ ,  $l = Mr^2 \dot{\phi}$

$\frac{dr}{dt} = \frac{dr}{d\phi} \frac{l}{Mr^2} \rightarrow$  substitute into total energy  $E$

$$\Rightarrow E = \frac{1}{2} M \left( \frac{dr}{d\phi} \right)^2 \frac{l^2}{M^2 r^4} + \frac{l^2}{2Mr^2} + V(r)$$

$$\left( \frac{dr}{d\phi} \right)^2 = \sqrt{\frac{2}{M} (E - V_{\text{eff}}(r)) \cdot \left( \frac{Mr^2}{l^2} \right)^2} = \frac{Mr^2}{l^2} \sqrt{\frac{2}{M} [E - V_{\text{eff}}(r)]}$$

$$d\phi = \pm \frac{l}{M} \int \frac{dr}{r^2 \sqrt{\frac{2}{M} [E - V_{\text{eff}}]}}$$

$$\int_{\phi=\pi}^{\phi_m} d\phi = \pm \frac{l}{\sqrt{2M}} \int_{r=\infty}^{r_{\text{min}}} \frac{dr}{r^2 \sqrt{E - \frac{l^2}{2Mr^2} - V}}$$

To obtain general expression  $\theta(b)$ , need to relate  $(E, l) \rightarrow b$   
and also use substitution:  $U = \frac{1}{r}$   $dU = -\frac{dr}{r^2}$

At  $r \rightarrow \infty$ :

$$l = M v_{\infty} b \rightarrow M b \left( \frac{2E}{M} \right)^{1/2} = b \sqrt{2ME} \Rightarrow \underline{l^2 = b^2 2ME}$$

$$E = \frac{1}{2} M v_{\infty}^2 \rightarrow v_{\infty}^2 = \frac{2E}{M}$$

Making the substitution of  $U = \frac{1}{r}$ ,  $dU = -\frac{dr}{r^2}$ ,  $l^2 = b^2 2ME$   
we obtain  $\Rightarrow$  (back page)

6) PART A continued...  $U_0 = \frac{1}{2} v_{\text{rel}}$

$$\phi_m - \pi = \frac{-b\sqrt{2ME}}{\sqrt{2M}} \int_0^{\infty} \frac{dU}{\sqrt{E - \frac{b^2 2ME U^2}{2M} - V}}$$

$$= - \int \sqrt{\frac{b^2 2ME}{(E - b^2 EU^2 - V)}} dU$$

$$= - \int \sqrt{\frac{Eb^2}{E(1 - b^2 U^2 - V/E)}} dU$$

$$\phi_m - \pi = -b \int \frac{dU}{[(1 - V/E) - b^2 U^2]^{1/2}}$$

↓

from geometry

$$\pi = \theta + 2(\pi - \phi_m)$$

$$= \theta + 2\pi - 2\phi_m$$

$$2\phi_m = \frac{\theta}{2} + \frac{\pi}{2}$$

$$\frac{\theta}{2} + \frac{\pi}{2} - \pi = -b \int \frac{dU}{[(1 - V/E) - b^2 U^2]^{1/2}}$$

$$- \frac{\pi}{2}$$

$$\theta - \pi = -2b \int \frac{dU}{[(1 - V/E) - b^2 U^2]^{1/2}}$$

$$+ \pi$$

$$\theta(b) = \pi - 2b \int_0^{U_0} \frac{dU}{[(1 - V/E) - b^2 U^2]^{1/2}}$$

6) PART B

$$\theta(b) = \pi - 2b \int_0^{u_0} \frac{du}{\left[1 - b^2 u^2 - \frac{V(u)}{E}\right]^{1/2}}$$

$$V(r) = \frac{k}{r} \quad (u \equiv \frac{1}{r})$$

$$V(u) = ku$$

$u_0 \equiv \frac{1}{r_{\min}}$ ,  $r_{\min} \rightarrow$  distance of closest approach, hence a turning point  $\Rightarrow \frac{dr}{dt} = \frac{dr}{d\phi} \frac{d\phi}{dt} = 0$

from part A:  $\left. \frac{dr}{dt} \right|_{r_{\min}} = \frac{Mr^2}{L} \sqrt{\frac{2}{u} \left[ E - \frac{L^2}{2mr^2} - V(r) \right]} \cdot \frac{1}{Mr^2} = 0$

use  $L^2 \equiv b^2 2uE$ ,  $\Rightarrow E - b^2 Eu^2 - V(u) = 0$

factor out  $E \Rightarrow E(1 - b^2 u^2 - \frac{k}{E}u) = 0$

$$\Rightarrow \underbrace{-b^2 u^2}_{A u^2} - \underbrace{\frac{k}{E}u}_{B u} + \underbrace{1}_{C} = 0 \quad \left\{ \begin{array}{l} u_0 = \frac{-B \pm \sqrt{B^2 - 4AC}}{2A} \end{array} \right.$$

Since  $u_0 > 0$  ( $r_{\min} > 0$ ) and  $B < 0$ ,  $A < 0$ , we choose "+" so that  $u_0$  comes out positive.

$$u_0 = \frac{-B - \sqrt{B^2 - 4AC}}{2A}$$

To solve the integral, we may re-write as quadratic expression:

$$1 - b^2 u^2 - \frac{V(u)}{E} \rightarrow \underbrace{-b^2 u^2}_{A u^2} - \underbrace{\frac{k}{E}u}_{B u} + \underbrace{1}_{C}$$

$$\Rightarrow \theta(b) = \pi - 2b \int_0^{u_0} \frac{du}{[A u^2 + B u + C]^{1/2}}, \quad \text{where } \begin{array}{l} A \equiv -b^2 \\ B \equiv -\frac{k}{E} \\ C \equiv 1 \end{array}$$

$$\Rightarrow \int_0^{u_0} \frac{du}{[A u^2 + B u + C]^{1/2}} = \frac{1}{\sqrt{-A}} \sin^{-1} \left( \frac{-2A u - B}{\sqrt{B^2 - 4AC}} \right) \Big|_0^{u_0}$$

Evaluating the integral at the limits:

$$\frac{1}{\sqrt{-A}} \left[ \sin^{-1} \left( \frac{-2Au_0 - B}{\sqrt{B^2 - 4A}} \right) - \sin^{-1} \left( \frac{-B}{\sqrt{B^2 - 4A}} \right) \right]$$

Substituting  $u_0$ , we obtain:

$$\frac{1}{\sqrt{-A}} \left[ \sin^{-1} \left( \frac{-2A \left( \frac{-B - \sqrt{B^2 - 4A}}{2A} \right) - B}{\sqrt{B^2 - 4A}} \right) - \sin^{-1} \left( \frac{-B}{\sqrt{B^2 - 4A}} \right) \right]$$

$$\Rightarrow \frac{1}{\sqrt{-A}} \left[ \sin^{-1} \left( \frac{B + \sqrt{B^2 - 4A} - B}{\sqrt{B^2 - 4A}} \right) - \text{" " } \right]$$

$$\Rightarrow \frac{1}{\sqrt{-A}} \left[ \underbrace{\sin^{-1}(1)}_{\pi/2} - \sin^{-1} \left( \frac{-B}{\sqrt{B^2 - 4A}} \right) \right]$$

Substituting this result back into  $\theta(b)$  expression:  
Recall:  $A \equiv -b^2$ ,  $B \equiv -k/E$ ,  $C \equiv 1$

$$\begin{aligned} \theta(b) &= \pi - 2 \left( \frac{1}{\sqrt{b^2}} \left[ \frac{\pi}{2} - \sin^{-1} \left( \frac{k/E}{\sqrt{(k/E)^2 + 4b^2}} \right) \right] \right) \\ &= \pi - 2 \left( \frac{\pi}{2} - \sin^{-1} \left( \text{" " } \right) \right) \end{aligned}$$

$$\boxed{\theta(b) = 2 \sin^{-1} \left( \frac{k/E}{\sqrt{(k/E)^2 + 4b^2}} \right)}$$

Now solving for  $b(\theta)$ :

$$\Rightarrow \sin\left(\frac{\theta}{2}\right) = \frac{k/E}{\sqrt{(k/E)^2 + 4b^2}} \Rightarrow \sqrt{(k/E)^2 + 4b^2} = \frac{k/E}{\sin(\theta/2)}$$

$$\Rightarrow (k/E)^2 + 4b^2 = (k/E)^2 \frac{1}{\sin^2(\theta/2)} = (k/E)^2 \csc^2(\theta/2)$$

$$\Rightarrow 4b^2 = (k/E)^2 \csc^2(\theta/2) - (k/E)^2$$

$$= (k/E)^2 \left[ \csc^2(\theta/2) - 1 \right] = (k/E)^2 \cot^2(\theta/2)$$

$$\Rightarrow b^2 = \left(\frac{k}{2E}\right)^2 \cot^2(\theta/2) \Rightarrow \boxed{b(\theta) = \frac{k}{2E} \cot\left(\frac{\theta}{2}\right)}$$

To re-derive  $\left(\frac{d\Gamma}{d\Omega}\right)_{\text{rederived}}$ ,

recall:  $\frac{d\Gamma}{d\Omega} = \frac{b}{\sin\theta} \left| \frac{db}{d\theta} \right|$

$$\frac{db}{d\theta} = -\frac{k}{2E} \csc^2\left(\frac{\theta}{2}\right) \cdot \frac{1}{2} \Rightarrow \boxed{\frac{db}{d\theta} = -\frac{k}{4E} \csc^2\left(\frac{\theta}{2}\right)}$$

Now substituting  $b(\theta)$  and  $\frac{db}{d\theta}$  into  $\frac{d\Gamma}{d\Omega} = \frac{b}{\sin\theta} \left| \frac{db}{d\theta} \right|$

$$\Rightarrow \frac{d\Gamma}{d\Omega} = \frac{k}{2E} \frac{\cos(\theta/2)}{\sin(\theta/2) \sin\theta} \cdot \left| -\frac{k}{4E} \frac{1}{\sin^2(\theta/2)} \right|$$

$$= \frac{k}{2E} \frac{\cos(\theta/2)}{\sin(\theta/2) \left( 2 \sin(\theta/2) \cos(\theta/2) \right)} \cdot \left| -\frac{k}{4E} \frac{1}{\sin^2 \theta/2} \right|$$

$$\frac{d\sigma}{d\Omega} = \Rightarrow \frac{k^2}{16E^2 \sin^4(\theta/2)}, \quad \text{Recall: } k = zZe^2 > 0$$

$$\Rightarrow \boxed{\frac{d\sigma}{d\Omega} = \left( \frac{zZe^2}{4E \sin^2(\theta/2)} \right)^2}$$

# 7) PART A

Start from the general form of  $b(\theta)$  for scattering off a repulsive potential  $\frac{\gamma}{r}$ .

$$\Theta(b) = \pi - 2b \int_0^{u_0} \frac{du}{[(1 - V/E) - b^2 u^2]^{1/2}} \quad \begin{cases} u_0 \equiv 1/r_{\min.} \\ u = 1/r \\ du = -1/r^2 dr \end{cases}$$

→ determine the upper limit,  $u_0$ :

Recall  $\left. \frac{dr}{dt} \right|_{r_{\min}} = \sqrt{\frac{2}{\mu}} \sqrt{E - \underbrace{\frac{l^2}{2\mu r_{\min}^2}}_{\text{Recall: } l^2 \equiv b^2 2\mu E} - V(r_{\min})} = 0$

$$\Rightarrow E - \frac{b^2 E}{r_{\min}^2} - \frac{\gamma}{r_{\min}} = 0$$

$$\Rightarrow E - b^2 E u_0^2 - \gamma u_0 = 0 \Rightarrow E(1 - b^2 u_0^2 - \frac{\gamma u_0}{E}) = 0$$

$$\Rightarrow -(b^2 + \frac{\gamma}{E}) u_0^2 + 1 = 0$$

$$\underbrace{A u_0^2 + B u_0 + C = 0}_{\substack{\downarrow \\ 0}}$$

$$u_0 = (-B \pm \sqrt{B^2 - 4AC}) / (2A)$$

$$= \pm \sqrt{4(b^2 + \frac{\gamma}{E})} / (-2(b^2 + \frac{\gamma}{E}))$$

$$u_0 = \boxed{\pm \frac{1}{\sqrt{b^2 + \frac{\gamma}{E}}}}$$

Since  $u_0 \equiv 1/r_{\min}$

$$r_{\min} \geq 0 \Rightarrow u_0 \geq 0$$

hence, we select ONLY "+"

→ Solve the integral part:

$$\int_0^{u_0} \frac{du}{\left[1 - b^2 u^2 - \frac{V(u)}{E}\right]^{1/2}}$$

$$V(r) = \frac{\gamma}{r^2} \rightarrow V(u) = \gamma u^2 \quad (u^2 \equiv \frac{1}{r^2})$$

$$\Rightarrow \int_0^{u_0} \frac{du}{\left[1 - b^2 u^2 - \frac{\gamma u^2}{E}\right]^{1/2}} = \int_0^{u_0} \frac{du}{\left[1 - u^2 \left(b^2 + \frac{\gamma}{E}\right)\right]^{1/2}}$$

$$\Rightarrow \int_0^{u_0} \frac{du}{\left[1 - \frac{u^2}{u_0^2}\right]^{1/2}} = u_0 \sin^{-1}\left(\frac{u}{u_0}\right) \Big|_0^{u_0} \quad \text{Recall: } u_0^2 \equiv \frac{1}{\left(b^2 + \frac{\gamma}{E}\right)}$$

$$\Rightarrow u_0 \left[ \underbrace{\sin^{-1}(1)}_{\pi/2} - \underbrace{\sin^{-1}(0)}_{=0} \right] = u_0 \cdot \frac{\pi}{2}$$

→ Now, solve for  $b(\theta)$ :

$$\theta(b) = \pi - 2b \cdot \frac{\pi}{2} u_0 = \pi(1 - bu_0)$$

$$\theta = \pi - \pi bu_0 \Rightarrow \frac{\pi bu_0}{\pi} = \frac{\pi - \theta}{\pi}$$

$$(bu_0)^2 = \frac{(\pi - \theta)^2}{\pi^2} \Rightarrow \frac{b^2}{\left(b^2 + \frac{\gamma}{E}\right)} = \frac{(\pi - \theta)^2}{\pi^2}$$

$$\Rightarrow \frac{1}{1 + \frac{\gamma}{b^2 E}} = \frac{(\pi - \theta)^2}{\pi^2} \Rightarrow 1 + \frac{\gamma}{b^2 E} = \frac{\pi^2}{(\pi - \theta)^2} - 1$$

$$\Rightarrow \frac{\gamma}{b^2 E} = \frac{\pi^2 - (\pi - \theta)^2}{(\pi - \theta)^2} \Rightarrow \frac{\gamma}{b^2 E} = \frac{\pi^2 - (\pi^2 - 2\pi\theta + \theta^2)}{(\pi - \theta)^2}$$

$$\frac{\gamma}{b^2 E} = \frac{2\pi\theta - \theta^2}{(\pi - \theta)^2} \Rightarrow \boxed{b^2 = \frac{\gamma}{E} \frac{(\pi - \theta)^2}{\theta(2\pi - \theta)}}$$

## PART B

$$\frac{d\sigma}{d\Omega} = \frac{b}{\sin\theta} \left| \frac{db}{d\theta} \right|$$

$$\begin{aligned} \frac{d(b^2)}{d\theta} &= 2b \left| \frac{db}{d\theta} \right| = \frac{\gamma}{E} \left\{ \frac{-2(\pi - \theta)}{(2\pi\theta - \theta^2)} - \frac{(\pi - \theta)^2}{(2\pi\theta - \theta^2)^2} \cdot [2\pi - 2\theta] \right\} \\ &= \frac{\gamma}{E} \left\{ \frac{-2(\pi - \theta)(2\pi\theta - \theta^2) - (\pi - \theta)^2(\pi - \theta) \cdot 2}{(2\pi\theta - \theta^2)^2} \right\} \end{aligned}$$

$$\begin{aligned} \Rightarrow -\frac{2\gamma}{E} \left\{ \frac{2\pi^2\theta - \pi\theta^2 - 2\pi\theta^2 + \theta^3 + (\pi^2 - 2\pi\theta + \theta^2)(\pi - \theta)}{(2\pi\theta - \theta^2)^2} \right\} \\ - \frac{2\gamma}{E} \left\{ \frac{2\pi^2\theta - \pi\theta^2 - 2\pi\theta^2 + \theta^3 + \pi^3 - \pi^2\theta - 2\pi^2\theta + 2\pi\theta^2 + \theta^2\pi - \theta^3}{\theta^2(2\pi - \theta)^2} \right\} \end{aligned}$$

$$\Rightarrow \cancel{2}b \left| \frac{db}{d\theta} \right| = -\cancel{2}\frac{\gamma}{E} \left\{ \frac{\pi^2(\pi - \theta)}{\theta^2(2\pi - \theta)^2} \right\} = -\frac{2\gamma}{E}$$

Now substitute  $b = \left(\frac{\gamma}{E}\right)^{1/2} \cdot \frac{(\pi - \theta)}{[\theta(2\pi - \theta)]^{1/2}}$  into the above expression.

$$\frac{db}{d\theta} = \frac{[\theta(2\pi - \theta)]^{1/2}}{\left(\frac{\gamma}{E}\right)^{1/2} \cdot (\pi - \theta)} \cdot (\gamma/E) \cdot \frac{\pi^2(\pi - \theta)}{\theta^2(2\pi - \theta)^2}$$

$$= \left(\frac{\gamma}{E}\right)^{1/2} [\theta(2\pi - \theta)]^{(1/2 - 2)} \cdot \pi^2$$

$$\frac{db}{d\theta} = \left(\frac{\gamma}{E}\right)^{1/2} \frac{\pi^2}{[\theta(2\pi - \theta)]^{3/2}}$$

$$\frac{d\sigma}{d\Omega} = \frac{b}{\sin\theta} \left| \frac{db}{d\theta} \right|$$

$$= \frac{1}{\sin\theta} \cdot \left( \frac{\gamma}{E} \right)^{1/2} \frac{(\pi - \theta)}{[\theta(2\pi - \theta)]^{1/2}} \cdot \left( \frac{\gamma}{E} \right)^{1/2} \frac{\pi^2}{[\theta(2\pi - \theta)]^{3/2}}$$

$$= \frac{\gamma \pi^2}{E \sin\theta} \frac{(\pi - \theta)}{\theta^2 (2\pi - \theta)^2} \quad \checkmark$$

$$\Rightarrow \boxed{\frac{d\sigma}{d\Omega} = \frac{\gamma \pi^2}{E \sin\theta} \frac{\pi - \theta}{\theta^2 (2\pi - \theta)^2}}$$